THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · NCERT-TYPE UNIT TEST SET 1

Chapters 1–4: **Real Numbers, Polynomials, Pair of Linear Equations in Two Variables, Quadratic Equations** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The exponent of 3 in the prime factorisation of 2250 is: 1

(a) 4 **(b)** 3 **(c)** 2 **(d)** 5

2. The number of zeroes of a polynomial whose graph cuts the x-axis at 1 point(s) is: 1

(a) 3 **(b)** 0 **(c)** 1 **(d)** 2

3. The pair of equations $3x + 2y = 5$ and $9x + 6y = 15$ is: 1

(a) exactly two solutions
(b) no solution (parallel lines)
(c) exactly one solution (intersecting lines)
(d) infinitely many solutions (coincident lines)

4. For the equation $x^2 - 7x + 12 = 0$, the sum of the roots is: 1

(a) -7 **(b)** 7 **(c)** 12 **(d)** -5

5. If HCF(306, 657) = 9, then LCM(306, 657) is: 1

(a) 22338 **(b)** 963 **(c)** 2754 **(d)** 201042

6. If one zero of $x^2 - 7x + 10$ is 2, the other zero is: 1

(a) -5 **(b)** 9 **(c)** 5 **(d)** 10

7. The solution of $x + y = 8$, $x - y = 2$ has $x =$ 1

(a) 3 **(b)** 5 **(c)** -5 **(d)** 8

8. For $x^2 - 6x + k = 0$ to have equal roots, $k =$ 1

(a) 9 **(b)** 36 **(c)** 3 **(d)** 6

9. The product of a non-zero rational and an irrational number such as $2\sqrt{3}$ is: 1

(a) always irrational **(b)** always an integer
(c) always rational **(d)** sometimes rational

10. A quadratic polynomial whose zeroes are 3 and 4 is: 1

(a) $x^2 - 7x - 12$ **(b)** $x^2 + 7x + 12$ **(c)** $x^2 + 7x - 12$ **(d)** $x^2 - 7x + 12$

11. The solution of $2x + 3y = 12$, $5x + 4y = 23$ has $y =$ 1

(a) 3 **(b)** 2 **(c)** -2 **(d)** 5

12. If one root of $x^2 - 9x + k = 0$ is 2, then k is: 1

(a) 7 **(b)** 14 **(c)** 9 **(d)** 16

13. The decimal expansion of $\frac{13}{3125}$ terminates after how many places of decimals? 1

(a) 8 **(b)** 5 **(c)** 7 **(d)** 6

14. If α, β are zeroes of $x^2 - 9x + 14$, then $\alpha + \beta$ is: 1

(a) 14 **(b)** 9 **(c)** -9 **(d)** 10

15. For the pair $x + y = 10$ and $x - y = 2$, the value of xy is: 1

(a) 26 **(b)** 10 **(c)** 48 **(d)** 24

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Check whether 6^n can end with the digit 0 for any natural number n. 2

17. The graph of $y = p(x)$ cuts the x-axis at $x = -2$ and $x = 3$ only. What are the zeroes of $p(x)$, and what is its likely degree? 2

18. A man tells his daughter, "Seven years ago I was 7 times as old as you were. Three years from now I shall be 3 times as old as you will be." Form the pair of linear equations (do not solve). 2

19. Represent the situation as a quadratic equation: The product of two consecutive positive integers is 72. 2

20. Given that HCF(306,657)=9, find LCM(306,657). 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. Prove that $\sqrt{2}$ is irrational. 3

22. Find the zeroes of $x^2 - x - 6$ and verify the relationship between the zeroes and the coefficients. 3

23. On comparing the ratios, find whether the lines $5x - 4y = 8$ and $7x + 6y = 9$ intersect, are parallel or coincide. 3

24. Find the roots of $2x^2 - 7x + 3 = 0$ using the quadratic formula (if they exist). 3

OR

Show that $3 + 2\sqrt{5}$ is irrational, given that $\sqrt{5}$ is irrational. 3

25. If 2 is a zero of the polynomial $x^2 + kx + 6$, find k and the other zero. 3

OR

Solve by elimination method: $2x + 3y = 12$ and $5x + 4y = 23$. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. Prove that $\sqrt{17}$ is irrational. 5

OR

If α, β are the zeroes of $x^2 - 7x + 10$, form a quadratic polynomial whose zeroes are $\alpha + 1$ and $\beta + 1$. 5

27. A father is 8 times as old as his son. After 9 years the father will be 5 times as old as the son will be. Find their present ages. **5**

OR

A train travels 360 km at a uniform speed. If the speed had been 5 km/h more, it would have taken 1 hour less for the journey. Find the original speed. **5**

END OF QUESTION PAPER **Total: 50 Marks**

MASTERMIND